Video-based fatigue estimation for human-robot task allocation optimisation

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***Abstract*— Human-centric manufacturing paradigm requires the human-robot collaboration (HRC) system to place the well-being of workers at the centre of manufacturing processes. Hence, optimising human workers' fatigue during human-robot task allocation in an HRC system is crucial for human-centric manufacturing. A prerequisite for this is an objective assessment of human fatigue. However, existing fatigue assessment relies on wearing clunky contact sensors, which are not user-friendly on an unstructured manufacturing shop floor. To address this limitation, we propose a video-based fatigue estimation method in which the boundary-aware dual-stream MS-TCN algorithm is proposed to detect operation type and operation repetitions from the video, and then the detected results are input into a fatigue model to estimate the worker's fatigue. In addition, the estimated human fatigue from the video is used as a basis for a human-robot task allocation optimisation model. The optimisation objective is to minimise cycle time while constraining human fatigue within an acceptable range. The experiment results show the validity of the fatigue estimation method, the superiority of the operation segmentation algorithm, and the effectiveness of the human-robot task allocation optimisation model.**

I. INTRODUCTION

Human-robot collaboration (HRC) plays a crucial role in futuristic manufacturing systems, as it enables ultra-flexible automation in unstructured environments [1]. In HRC, it is essential to optimise human-robot task allocation with efficiency as the objective, considering the respective advantages of each [2], [3]. However, driven by the Industry 5.0 initiative [4], the manufacturing paradigm will shift from system-centric, focusing on efficiency and quality improvement and cost reduction, to human-centric, placing the well-being of industry workers at the centre of manufacturing processes [5]. Therefore, towards human-centric manufacturing, task allocation in HRC must prioritise the well-being of workers rather than emphasising efficiency above all else. Among the factors (e.g., physical, cognitive, and psychological health [5]) that affect the well-being of workers, muscular fatigue is a quantifiable indicator of the physical workload impact on workers' bodies [6]. For this reason, it is of great significance to optimise human workers' fatigue as well as system efficiency through human-robot task allocation.

Currently, the objective assessment of muscular fatigue mainly relies on the measurement of contact sensors, such as surface-Electromyography (s-EMG). Exhaustive research into s-EMG based fatigue estimation methods has been explored [7], [8]. However, the limitation of these methods in manufacturing scenarios is manifest, as it is not realistic to let every worker wear s-EMG sensors during work. Therefore, fatigue estimation methods based on non-contact sensors need to be explored, such as cameras. The existing research indicates that the worker's fatigue is related to their muscle strength, object weight factor, operation type and operation repetitions [8], and the advancement of video understanding technologies can enable the recognition of operation type and operation repetitions from the video [9], [10]. Therefore, in this paper, we developed a video-based fatigue estimation method and used the estimated fatigue in human-robot task allocation optimisation.

This paper is organised as follows. Section II provides a brief review of existing research on fatigue estimation methods and video understanding technologies. Section III outlines the methods for the proposed video-based fatigue estimation and human-robot task allocation optimisation. Section IV discusses the results of a case study used to validate the proposed methods. Lastly, Section V concludes this paper.

II. RELATED WORK

To propose a video-based fatigue estimation method, it is essential to review the existing fatigue estimation methods and analyse the fatigue impact factors that can be recognised from the video. Then, it is necessary to review the available video understanding technologies to extract these factors.

A. *Fatigue estimation in manufacturing*

Existing fatigue estimation methods can be classified into either subjective or objective assessments. Subjective assessment evaluates fatigue based on individuals' opinions or preferences [11], [12], which vary significantly between individuals with differing perceptions. Consequently, such methods lack robustness and introduce personal bias when applied to dynamic applications such as human-robot task allocation. To address this limitation, researchers have tried objective assessment methods which directly measure physical attributes to infer fatigue accumulation over consistent muscle force generation. One such method named learning-forgetting-fatigue-recovery (LFFR) [13] estimates fatigue

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accumulation based on the assumption of the maximum endurance time. However, this method fails to identify localised fatigue which is responsible for short-term workplace injuries and long-term worker health risks and appears more commonly in manufacturing assembly due to the flexible but repetitive nature of tasks. Localised fatigue accumulation can be estimated from myoelectric signal changes measured from surface-Electromyography (s-EMG) sensors during repetitive operations until localised failure. A spectral shift to lower frequencies and an increased dynamic range of the myoelectric signal are observed as muscle fibre conduction velocity (MCV) decreases due to muscle exhaustion, where variations in myoelectric signal properties can be used as fatigue indicators [14]. Although this method is objective, the requirement of wearing a clunky s-EMG sensing system limits its application in real industrial assembly scenarios. To eliminate the reliance on contact sensors for the objective assessment of worker's fatigue, [8] derived and validated a model which can estimate the fatigue impact of a repeated operation based on the operation type, object weight factor, operation repetitions and the worker's muscle strength.

B. Video understanding

Action segmentation techniques in video understanding can be leveraged to extract operation types and operation repetitions from videos. Action segmentation aims at temporally locating and recognising human action segments (constituting by consecutive frames with the same action labels) in long untrimmed videos [15]. The early action segmentation approaches are based on a sliding window [16], [17] or the state change of objects [18], [19] to classify the action within a window or determine the action boundary, but these approaches cannot capture the context over long-time videos [10], which limits the performance. To tackle this problem, researchers have tried to employ a temporal model over a frame-wise classifier [20], [21]. However, temporal models are slow because they can only predict the action of each frame sequentially, not parallelly. Inspired by the success of temporal convolutional network (TCN) in speech synthesis, Lea et al. [22] introduced TCN into action segmentation. TCN can capture long-range dependencies with higher efficiency than the temporal models. Most current state-of-the-art action segmentation algorithms are developed based on TCN. The most representative work is MS-TCN [10], which stacks several TCN modules to refine the prediction stage-by-stage, and some subsequent studies have been done to improve MS-TCN [23], [24]. However, the above work cannot distinguish the boundary between continuously repeated actions because the algorithms are designed to predict the action type of each frame and then aggregate the continuous frames of the same action into an action segment. However, distinguishing continuously repeated actions to obtain the number of repetitions is essential for estimating fatigue. To address this problem, we designed a boundary-aware dual-stream MS-TCN network, one stream for predicting the end frame of the action, and another stream for predicting the action type of each frame, to recognise the continuously repeated actions and count the repetitions.

III. METHOD

This section introduces the video-based fatigue estimation framework for human-robot task allocation optimisation and its detailed methods.

A. Overall framework

Fig. 1 presents the overall framework of video-based fatigue estimation for human-robot task allocation optimisation. It contains three processes: operation segmentation, fatigue estimation, and task allocation optimisation. First, the operation segmentation takes videos as input and outputs the operation segments and the repetitions for each operation. Second, the worker's fatigue is estimated through the outputs of operation segmentation, the worker's muscle strength and the object weight factor. Finally, considering the worker's fatigue, the execution time of human and robot for each operation and the availability of human and robot, the human-robot task allocation is optimised for minimising the cycle time while ensuring the worker's fatigue is under an acceptable level.

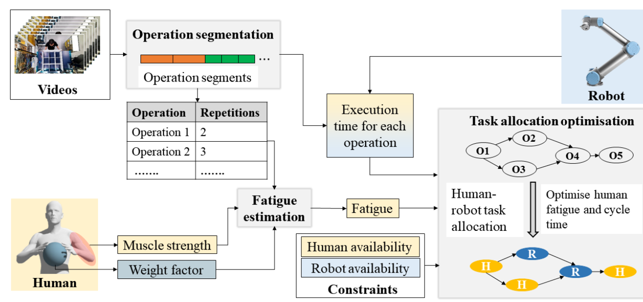

Fig. 1. Video-based fatigue estimation framework for human-robot task allocation optimisation.

B. Operation segmentation

To address the problem of separating continuously repeated operations, we propose the boundary-aware dual-stream MS-TCN, as shown in Fig. 2. It contains two streams: operation prediction stream (OPS) and boundary prediction stream (BPS). First, I3D features [9] for each frame of the input video are extracted as $X = x_1, ..., x_T \in R^{T \times D}$, where T denotes the number of frames of the input video and D denotes the dimension of the feature. Then, the two streams take X as the input and output frame-wise operation predictions, $O = [o_1, ..., o_T] \in \{0,1\}^{T \times C}$, and boundary predictions, $B = [b_1, ..., b_T] \in \{0,1\}^{T}$. Finally, boundary predictions are used to refine the operation predictions to get the final operation segments. These two streams are trained separately.

In the OPS stream, we use a multi-stage temporal convolutional network (MS-TCN) [10]. TCN block can operate on the full temporal resolution of the videos and capture long-term dependencies. With multi-stage architecture, the prediction results can be refined stage-by-stage. The output layer of the operation prediction stream is a 1D convolutional layer followed by a SoftMax layer to predict frame-wise operation class probabilities $Y = [y_1, ..., y_T] \in (0,1)^{T\times C}$, where C is the number of operation classes. Referring to [10], the loss function is a combination of a classification loss and a truncated mean squared error (T-MSE). It can be formulated as:

$$\mathcal{L}_{ops} = \mathcal{L}_{cls} + \lambda \mathcal{L}_{T-MSE}; \tag{1}$$

$$\mathcal{L}_{cls} = \frac{1}{T}\sum_{t} -log(y_{t,c}); \tag{2}$$

$$\mathcal{L}_{T-MSE} = \frac{1}{TC}\sum_{t,c} \widetilde{\Delta}^2_{t,c}; \tag{3}$$

$$\widetilde{\Delta}_{t,c} = \begin{cases} \Delta_{t,c}, & \Delta_{t,c} \le thresh, \\ thresh, & \text{otherwise}; \end{cases} \tag{4}$$

$$\Delta_{t,c} = |\log y_{t,c} - \log y_{t-1,c}|; \tag{5}$$

where $y_{t,c}$ is the probability of class c at time t. An argmax function is applied to map Y to O.

The BPS has the same network architecture as the OPS, but the output is the frame-wise boundary probabilities $P = [p_1, ..., p_T] \in (0,1)^T$. The loss function $\mathcal{L}_{bps}$ is obtained by replacing the $y_{t,c}$ in $\mathcal{L}_{ops}$ with p_t, the probability of boundary at time t. An argmax function is applied to map P to B.

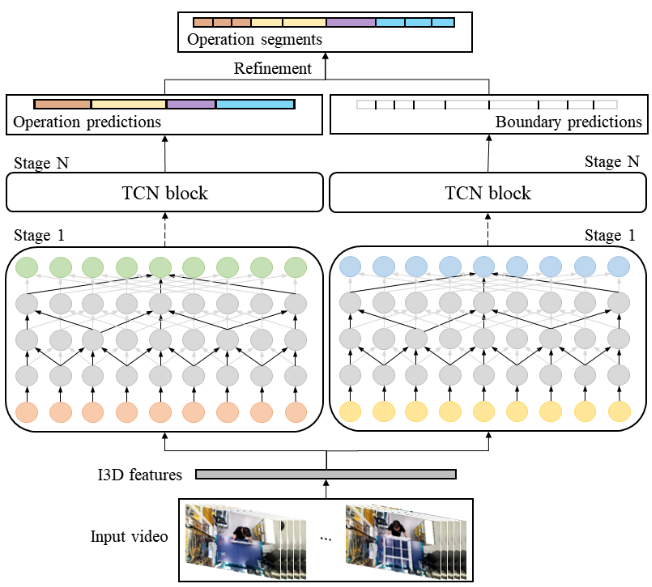

Fig. 2. Overview of the boundary-aware dual-stream MS-TCN network.

C. Fatigue estimation

In [8], it is assumed that any assembly operation involves both fatigue-inducing motions, such as manipulating an object with a significant weight factor, and non-fatigue-inducing motions, such as reaching for a part. Since the minuscule recovery periods (non-fatigue inducing motions) create negligible effects on fatigue, the fatigue can be assumed only related to the operation type o, object weight factor w, number of repetitions n and the worker's muscle strength s. The object weight factor w is the relative measure of an object's weight to a worker's muscle strength s. The muscle strength is the maximum weight of the target muscle (bicep brachii muscle in this paper) can carry under isometric conditions. It can be determined through the experiments described in [8]. Through empirical experiments, [8] confirmed a linear relationship between fatigue accumulation and repetitive operations n_i of a task i. Fatigue response F_i of a task i can be formulated as:

$$F_i = A_{o,w}\ n_i, \quad s.t. \quad n_i \le N, \tag{6}$$

where $A_{o,w}$ is the fatigue parameter of o and w combination, and n_i is the number of repetitions. The $A_{o,w}$ is found experimentally by determining the maximum number of repetitions N that results 100% fatigue (unable to execute another repetition) such that there is a consistent linear increase between fatigue accumulation and repetitions. The detailed determination method of $A_{o,w}$ and w for an operation type and s for a particular worker can be found in [8].

In LLFR model [13], the residual fatigue of a worker after a period of rest after the same task is modelled. Based on this, we propose the residual fatigue after a period of rest as:

$$R_i = F_i\ e^{-\mu\tau_i}, \tag{7}$$

where τ_i is the rest time after performing task i and μ is the recovery parameter determined by factoring the time it takes to go from 100% fatigue to completely recovered (0% fatigue).

D. Task allocation optimisation

In an HRC assembly, human workers and robots work in a shared space on independent or collaborative tasks for assembling a common product. A product can be completed by a sequence of tasks. Optimally allocating tasks between agents (human or robot) allows tasks to be offloaded from the human worker onto the robot to provide rest while maintaining production efficiency. We propose a human-robot task allocation optimisation model that allocates tasks to humans and robots based on their availability and capability to minimise total cycle time with an acceptable maximum human fatigue.

Consider a job contains a sequence of I tasks, and two agents, including a human worker h and a robot ro. The objective is to minimise the total cycle time of the job given the following equation:

$$\begin{aligned} Min: CT = &\sum_{i=1}^{I} X_i T_i^h + \\ &\sum_{i=1}^{I}(1-X_i)T_i^{ro} + \sum_{i=1}^{I}\tau_i \end{aligned} \tag{8}$$

$$s.t.$$

$$R_i = X_i F_i e^{-\mu\tau_i}, \quad i = 1 \tag{9}$$

$$R_i = X_i(R_{i-1} + F_i)e^{-\mu\tau_i} + (1 - X_i)R_{i-1}e^{-\mu T_i^{ro}}, \quad \forall i > 1 \tag{10}$$

$$R_{i-1} + F_i \leq FT, \tag{11}$$

where

$$X_i = \begin{cases} 1, & when\ task\ i\ assigned\ to\ h, \\ 0, & when\ task\ i\ assigned\ to\ ro; \end{cases} \tag{12}$$

T_i^h and T_i^{ro} are the execution time required by the human worker and the robot to complete the task i respectively; τ_i is the time of human rest after task i; R_i is the residual fatigue of the human worker after resting τ_i time after task i; FT is the fatigue threshold. Eq. (11) is to ensure the maximum fatigue of the human worker during the job is under an acceptable level.

With the above optimisation model, a genetic algorithm (GA) is designed to optimise human-robot task allocation. The pseudocode of GA is as follows:

Algorithm 1: GA for human-robot task allocation optimisation

```
Initialize population P_t of size nPop;
for each individual p_t in P_t do
    EvaluateFitness(p_t);
end
for MaxIt iterations do
    Using Tournament Selection:
    - Create Pool of size nPool from P_t;
    - Choose nTour individuals from Pool;
    - Select best parent;
    Crossover individuals to create nCrossover
      offspring;
    Mutate individuals to create nMutation
      offspring;
    for each individual p'_t in modified population P'_t
      do
        EvaluateFitness(p'_t);
    end
    P_{t+1} ← best nPop individuals from P'_t;
    t ← t + 1;
end
```

IV. Case study

A. Video-based fatigue estimation

We took a bookshelf assembly as a research case to verify the effectiveness of the proposed video-based fatigue estimation method. First, we repeatedly assembled the bookshelf 20 times to obtain 20 videos, then randomly selected 16 videos as the training set and the remaining four videos as the testing set. Each video contains nine operation types, and 64 operation segments, as shown in Table II. Finally, we compared the operation segmentation performance of MS-TCN and our boundary-aware dual-stream MS-TCN on this dataset, and the results are shown in Table IV-A. In our network, for each stream, we stack four stages, and each contains 10 dilated residual convolution layers. The dilation factor is doubled at each layer, and dropout rate is 0.5 for each layer. Each layer has 64 filters, and filter size is three. For the loss function, $\lambda = 0.15$ and $thresh = 4$.

TABLE I
Performance comparison between MS-TCN and ours.

	Frame-wise accuracy	Operation segments per video
MS-TCN	99.87%	26/64
Ours	99.97%	64/64

The frame-wise prediction accuracy in Table IV-A shows that both MS-TCN and our algorithm can effectively identify the operation type of each frame in a video. The two algorithms have a different number of operation segments because MS-TCN is unable to segment continuously repeated operations, as depicted in Fig. 3. On the other hand, our algorithm can segment such continuously repeated actions, which is crucial for estimating the worker's fatigue. Considering the continuously repeated operations, the ground truth of operation segments is 64. Therefore, the superiority of our algorithm is validated. While our algorithm demonstrated near-perfect performance in this experiment, it does not imply that it has no flaws. We attribute the superior performance to two limitations: (1) the dataset size is limited, and (2) each video contains the same operation sequence. The gap between our dataset and the real-world situation highlights the need for additional efforts to explore the practical application of this method in the future.

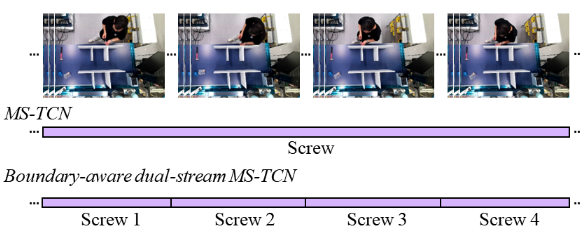

Fig. 3. The illustration of operation segmentation results of MS-TCN and Boundary-aware dual-stream MS-TCN.

After the above operation segmentation, we can obtain a table of operation types and operation repetitions in the video. Through eq. (6), fatigue corresponding to each operation in the video is obtained, as shown in Table II. The fatigue parameter for each operation is obtained through the method described in [8].

B. Human-robot task allocation optimisation

We present a sample task allocation application involving common HRC operations for assembling a bookshelf. The task order for the bookshelf assembly is shown in the precedence graph in Fig. 4. The agents' capabilities on each

TABLE II

DETAILED TASK INFORMATION FOR THE BOOKSHELF ASSEMBLY.

Task index	Operation type*	Repetitions	Fatigue param	Fatigue response	Human E-time (s)	Robot E-time (s)	HRC E-time (s)
1	MLP	1	8.33	8.33	22	29	-
2	MSP	4	4	16	34	45	-
3	SS	2	2	4	127	-	52
4	MLP	1	8.33	8.33	22	29	-
5	MSP	4	4	16	34	45	-
6	SS	2	2	4	127	-	52
7	MLP	1	8.33	8.33	22	29	-
8	MSP	4	4	16	34	45	-
9	SS	2	2	4	127	-	52
10	HD	4	5	20	31	-	-
11	SC	2	5	10	45	-	39
12	IC	1	10	10	39	-	-
13	HD	4	5	20	31	-	-
14	SC	2	5	10	45	-	39
15	IC	1	10	10	39	-	-
16	R1	1	8.33	8.33	21	13	-
17	MLP	1	8.33	8.33	22	29	-
18	HD	4	5	20	31	-	-
19	SC	2	5	10	45	-	39
20	IC	1	10	10	39	-	-
21	R2	1	10	10	25	15	-
22	MLP	1	5	5	22	29	-
23	SS	8	2	16	254	-	104
24	R3	1	11.1	11.1	31	19	-
25	MLP	1	8.33	8.33	22	29	-
26	SS	8	2	16	254	-	104

**MLP: moving a large plate; MSP: moving a small plate;*
SS: screwing a screw; HD: hammering a dowel;
SC: screwing a connector; IC: inserting a component; R1,2,3: rotating 1,2,3.

TABLE III

IMPLEMENTATION DETAILS OF THE GA.

nPop (number of population)	120
MaxIt (maximum iteration)	250
nCrossover	0.3 $\times$ nPop
nMutation	0.2 $\times$ nPop
Mu	0.05 $\times$ cell in chromosome
nPool	0.7 $\times$ nPop
nTour	4 $\times$ nPop

task are given in Fig. 4, where the capability has three categories: (1) both human and robot can complete the task; (2) only human can complete the task; and (3) the task requires both human and robot to collaborate. Detailed task information is provided in Table II.

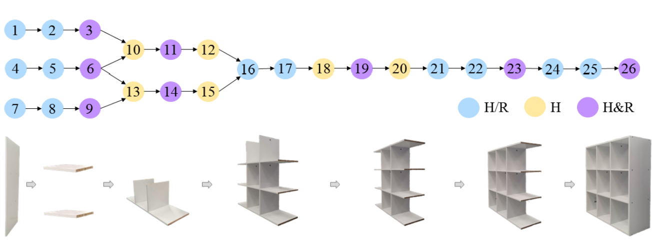

Fig. 4. Precedence graph for the bookshelf assembly.

Based on the precedence graph in Fig. 4 and the task details in Table II, the human-robot task allocation model can be formulated, and the fatigue threshold is set to 0.7. GA is implemented with the parameters shown in Table III.

After the optimisation, a resolution is obtained, as shown in Table IV. If task i is assigned to human, $X_i = 1$, otherwise, $X_i = 0$. τ_i is the time of rest after task i. In this resolution, the total cycle time is 1307s. This experiment proves the effectiveness of the optimisation model and algorithm.

TABLE IV

HUMAN-ROBOT TASK ALLOCATION OPTIMISATION RESULT.

Task i	1	2	3	4	5	6	7	8	9	10	11	12	13
X_i	1	0	1	1	0	1	1	0	1	1	0	1	1
τ_i	0	0	0	0	0	0	0	0	0	8	0	0	17
Task i	14	15	16	17	18	19	20	21	22	23	24	25	26
X_i	0	1	1	0	1	1	1	1	1	0	1	1	0
τ_i	0	0	0	0	0	9	0	0	0	0	0	0	0

V. CONCLUSIONS

This paper proposes a video-based fatigue estimation method, in which the boundary-aware dual-stream MS-TCN algorithm is proposed to recognise operation type and operation repetition from the video, and then the outputs of the algorithm are input into a fatigue model to estimate the worker's fatigue. This method can quantitatively estimate worker fatigue from video and thus can be integrated into downstream human-centric optimisation tasks. We also develop a human-robot task allocation optimisation model aiming at improving the well-being of workers and ensuring the efficiency of an HRC system. The case study verifies the validity of the fatigue estimation method, the superiority of the operation segmentation algorithm and the effectiveness of the human-robot task allocation optimisation method. To address the limitations of the paper, we plan to a) improve the video-based fatigue estimation method to be real-time online estimation, b) adapt the improved fatigue estimation method to real-time task scheduling, and c) consider parallel tasks and multiple agents in task scheduling.

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